

SimRail: a normal contact problem library in python. Implementation solution of the normal contact problem with the equivalent elastic method

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Abstract

This paper introduces SimRail, a Python library for solving the normal contact problem in railway dynamics. The library implements both the equivalent elastic contact method as implemented in SIMPACK and the originally based work of Piotrowski and Kik. SimRail implementation is detailed step by step, including wheel-rail positioning, coordinate interpolation, contact region identification, rotation to the normal direction, and parameter calculation. The step by step description is provided for the equivalent elastic method. Validation against SIMPACK under different rail inclinations (1:40, 1:30, 1:20) and a 50 kN Q force shows excellent agreement, with R^2 values up to 0.999 for penetration and 0.997 for normal force, and minimal errors in key metrics. SimRail is openly available on GitHub for research in railway vehicle dynamics.

1 Introduction

The normal contact problem is a fundamental aspect in railway dynamics and its solution is the key for accurate simulations. Several methods have been proposed, among which the equivalent elastic method. This method, as implemented in SIMPACK, has been described by Vollebregt in [1]. The equivalent elastic method is based on the work proposed by J. Piotrowski and W. Kik [2]. Two key aspects are kept from the work by J. Piotrowski and W. Kik. The first one is the normal force calculation. This is obtained from a prescribed virtual interpenetration of the wheel and the rail in the contact region and not the opposite. The second is that the length and shape in longitudinal direction of the contact patch are determined by the combination of the geometries of the wheel and rail in the interpenetration region. In contrast to the work by J. Piotrowski and W. Kik, the correction of the contact patch shape is not performed in the equivalent elastic method. Moreover, the virtual interpenetration distance is not decreased by the suggested factor of 0.55.

According to Vollebregt, the patch determined with J. Piotrowski and W. Kik approach is simplified in an elliptical contact patch where the width and length are obtained as in Figure 1. To retain part of the original contact patch shape information into the equivalent ellipse, the normal force is applied to the centre of gravity (centroid) of the non-elliptical contact patch. Once the width of the interpenetration section is found, the penetration used to determine the normal force is found by determining the penetration of a circle segment (or more properly the height of the circle segment) that has the same width and cross-section area as the actual

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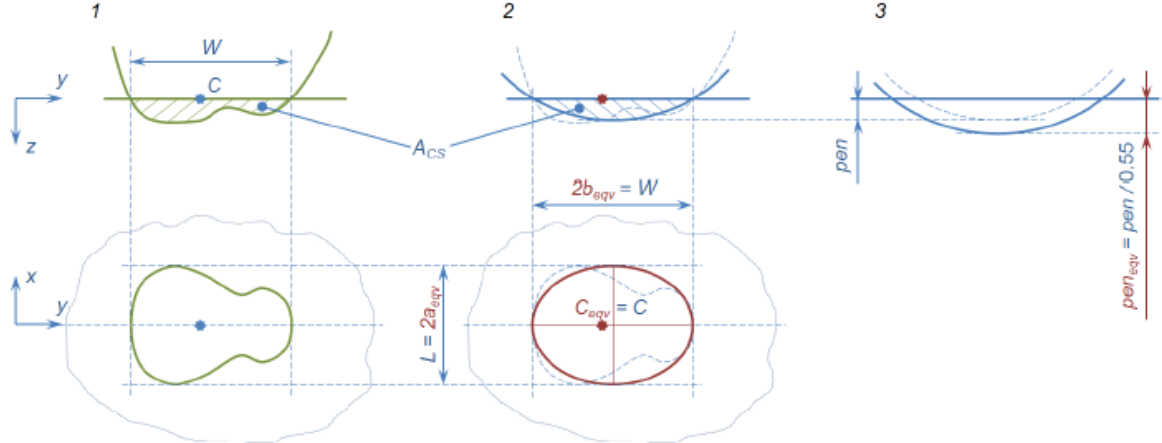


Figure 1 *Equivalent-elastic contact method. W = width, L = max. length. A_{cs} = interpenetration cross-section area. C = contact reference point, pen = penetration or approach, a, b = semi-axes. The subscript “eqv” means “equivalent”.*

Figure 1: Implementation of the contact patch search for the equivalent contact method (SIMPACK) [1].

interpenetration calculated with J. Piotrowski and W. Kik approach (Figure 1 (2 top)). The actual penetration (or approach) used to calculate the normal force is the penetration found with the equivalent circle segment increased by a factor $1/0.55$.

The goal of this paper is to reproduce the results of the equivalent elastic method as implemented in SIMPACK [1] and to provide a library that can be used to calculate the normal contact problem with the equivalent elastic method. The library is called **SimRail** and is available on GitHub (<https://github.com/RoccoLiberoGiossi/SimRail>). The library also include the normal contact problem calculation with the J. Piotrowski and W. Kik approach [2]. The library is written in python and can be used to calculate the normal contact problem for any wheel-rail pair.

What follows is a description of the implementation of the equivalent elastic method in SimRail. The implementation is described step by step, starting from the position of the wheel and rail in their location in the y-z plane, to the calculation of the normal force.

2 Practice of calculation

2.1 Position wheel and rail in their location in the y-z plane

In Figure 2 the wheel and rail location in the y-z plane for the right wheel-rail pair is shown as example considering semi-axis distance $b_0 = 0.75$, wheel profile S1002, rail profile UIC60, rail inclination 1:40, and gauge = 1.435 m. The z-axis direction is provided accordingly to the railway reference frame coordinates system. Generally there will be one or more intersections between the wheel and the rail. Those intersections are the regions where the contact patches will appear.

2.2 Interpolation of Wheel in Rail coordinates.

To simplify the calculation of the contact patch shape as function of the wheel and rail lateral coordinates, the wheel can be re-discretized in the rail lateral coordinates to obtained the section

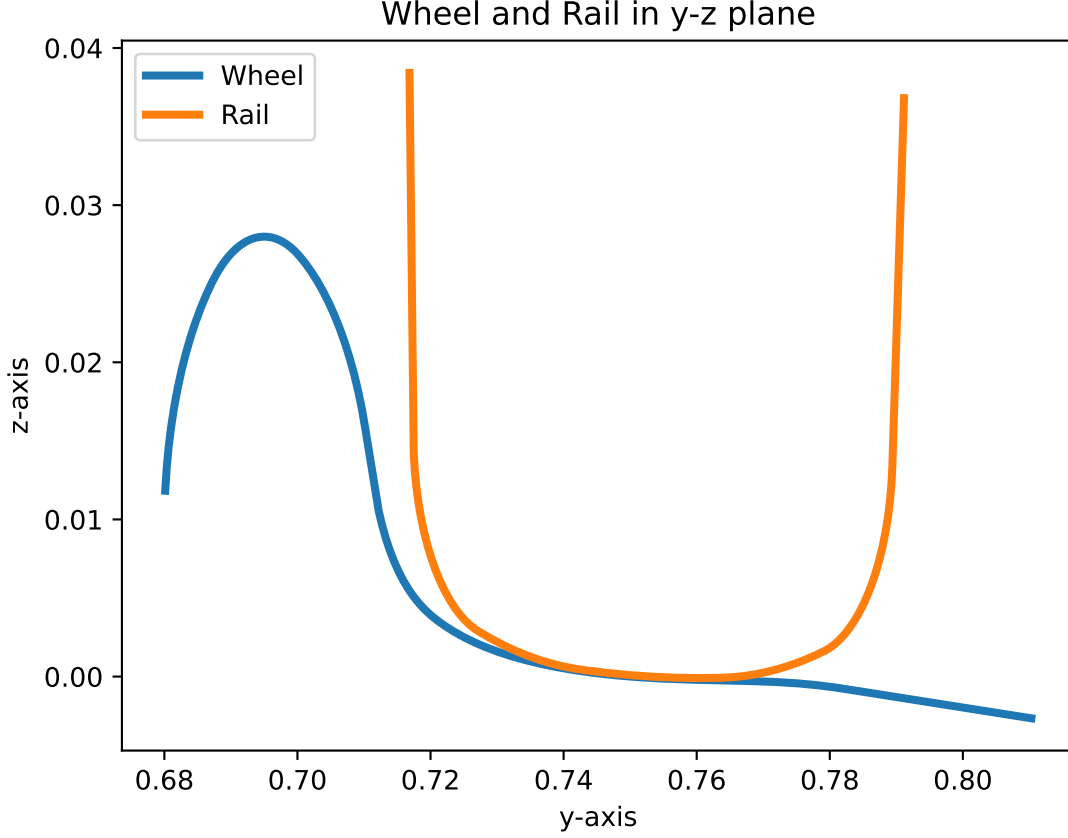


Figure 2: Wheel and rail location.

of the wheel that is “seen” by the rail in the relative location of the wheel and the rail, as in the example of Figure 2. In this way the difference in vertical direction between the wheel and rail profiles is “freed” from the lateral coordinate dependencies of the wheel and the rail giving the wheel and rail vertical coordinates function only of the rail lateral coordinates (Figure 3). The same can be done for the wheel radius. Thus, the wheel, rail, and wheel radius coordinates can be expressed,

$$y_{\text{wheel}}^{\text{rail}} = y_{\text{rail}}, \quad z_{\text{wheel}} = f(y_{\text{rail}}), \quad z_{\text{rail}} = g(y_{\text{rail}}), \quad r = h(y_{\text{rail}}), \quad (1)$$

where, $y_{\text{wheel}}^{\text{rail}}$ is the wheel lateral coordinate expressed in the rail coordinates, z_{wheel} is the wheel vertical coordinate, and z_{rail} is the rail vertical coordinate. The function f is the interpolation functions of the wheel over the rail coordinate, h is the interpolation functions of the wheel radius r over the rail coordinate, while the function g is the rail profile function.

2.3 Find the position

Now that the wheel and rail are both function of the rail lateral coordinates, it is possible to find the vector indexes where the difference between wheel and rail vertical coordinates vectors is greater than 0.

$$\text{Idx}_{\text{contact}} = [(z_{\text{wheel}} - z_{\text{rail}}) > 0] \quad (2)$$

This method is general in the sense that it is not restricted to one contact patch [2]. If in the resultant difference vector there are indexes that have more than one index distance, then there

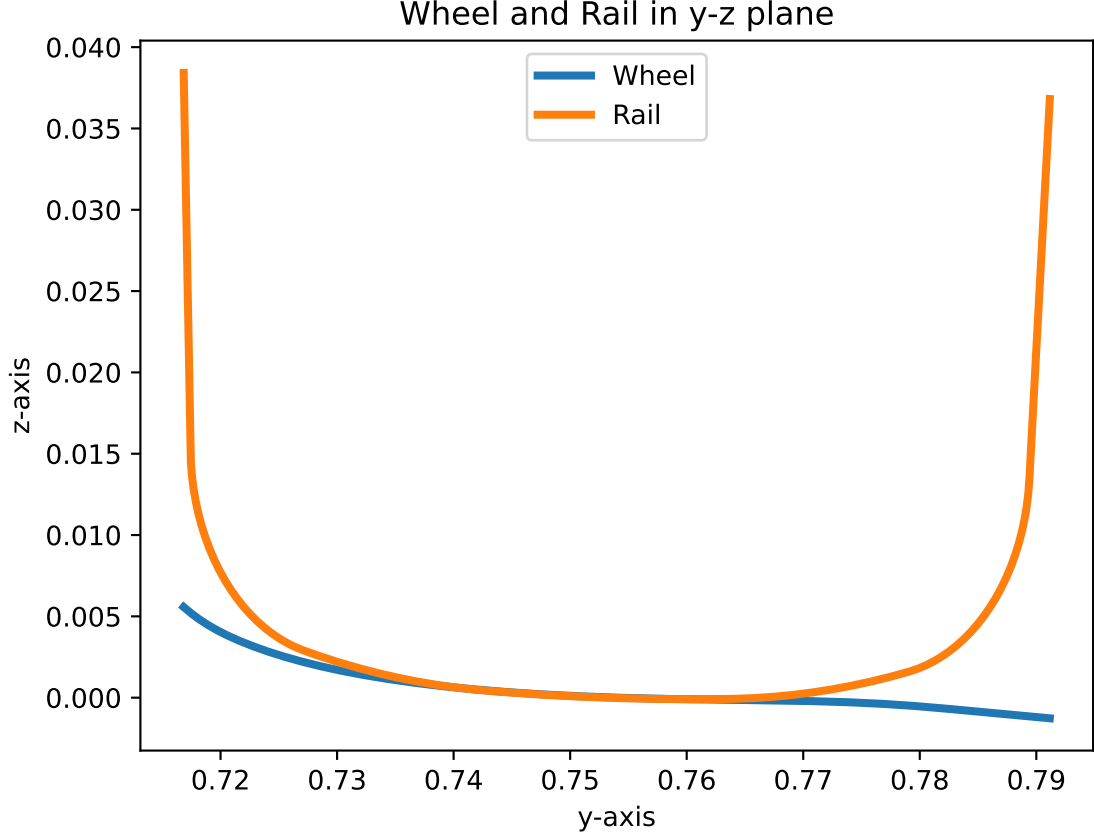


Figure 3: Wheel segment as function of the Rail coordinates

are more than one contact patch. As example for simplicity of explanation consider the contact indexes,

$$\begin{aligned} \text{Idx}_{\text{contact}} &= [0, 1, 2, 3, 4, 7, 8, 9, 10, 11] \rightarrow \\ \text{Idx}_{\text{contact},1} &= [0, 1, 2, 3, 4], \quad \text{Idx}_{\text{contact},2} = [7, 8, 9, 10, 11] \end{aligned} \quad (3)$$

2.4 Rotate to Normal direction

The identified intersecting regions need to be rotated in the normal direction to calculate the approach and the contact parameters. For each identified contact patch (k), find the contact rotation angle α ,

$$\alpha = -\arctg \left(\frac{z_{\text{rail}}(\text{Idx}_{\text{contact}_k}(\text{end})) - z_{\text{rail}}(\text{Idx}_{\text{contact}_k}(\text{ini}))}{y_{\text{rail}}(\text{Idx}_{\text{contact}_k}(\text{end})) - y_{\text{rail}}(\text{Idx}_{\text{contact}_k}(\text{ini}))} \right), \quad (4)$$

where, $y_{\text{rail}}, z_{\text{rail}}(\text{Idx}_{\text{contact}_k}(\text{end}))$ and $y_{\text{rail}}, z_{\text{rail}}(\text{Idx}_{\text{contact}_k}(\text{ini}))$ are the lateral y and vertical z coordinates of the rail profile in the contact region k at the last identified (end) respectively the first identified (ini) contact indexes. The rotation matrix R_α can now be computed,

$$R_\alpha = \begin{bmatrix} \cos(\alpha) & -\sin(\alpha) & 0 \\ \sin(\alpha) & \cos(\alpha) & 0 \\ 0 & 0 & 1 \end{bmatrix} \quad (5)$$

For each point in the contact patch region calculate the normal coordinates of wheel, rail, and wheel radius. To properly rotate each patch, a normalization is needed in order to have the origin coincide with the beginning of the coordinates vector.

$$\begin{bmatrix} y^{Normal} \\ z_{rail}^{Normal} \\ 0 \end{bmatrix} = R_\alpha \begin{bmatrix} y_{rail}(Idx_{contact_k}) - y_{rail}(Idx_{contact_k}(ini)) \\ z_{rail}(Idx_{contact_k}) - z_{rail}(Idx_{contact_k}(ini)) \\ 0 \end{bmatrix} \quad (6a)$$

$$\begin{bmatrix} y^{Normal} \\ z_{wheel}^{Normal} \\ 0 \end{bmatrix} = R_\alpha \begin{bmatrix} y_{rail}(Idx_{contact_k}) - y_{rail}(Idx_{contact_k}(ini)) \\ z_{wheel}(Idx_{contact_k}) - z_{wheel}(Idx_{contact_k}(ini)) \\ 0 \end{bmatrix} \quad (6b)$$

$$\begin{bmatrix} y^{Normal} \\ r^{Normal} \\ 0 \end{bmatrix} = R_\alpha \begin{bmatrix} y_{rail}(Idx_{contact_k}) - y_{rail}(Idx_{contact_k}(ini)) \\ r(Idx_{contact_k}) - r(Idx_{contact_k}(ini)) \\ 0 \end{bmatrix} \quad (6c)$$

where, y^{Normal} is the lateral coordinate vector in the normal direction, and z_{rail}^{Normal} , z_{wheel}^{Normal} , and r^{Normal} are the normal rail coordinate vector, the normal wheel coordinate vector, respectively the normal wheel radius coordinate vector. It is necessary to create an interpolation vector representing the discretization strips to solve the patch with. The default discretization used in SimRail is 58. To do so, re-discretize the y^{Normal} vector into a new one $\bar{y}^{Normal}$. This is useful for two aspects. The first being avoiding discretization problems due to the limited points number of the wheel and rail profiles. The second is to re-adjust the wheel, rail, and wheel radius to the same lateral normal direction. Interpolate now z_{rail}^{Normal} , z_{wheel}^{Normal} , and r^{Normal} over the $\bar{y}^{Normal}$ and remove the bias introduced to rotate to normal direction by reposition in the correct vertical location,

$$\bar{z}_{wheel}^{Normal} = f'(\bar{y}^{Normal}) + z_{wheel}(Idx_{contact_k}(ini)) \quad (7a)$$

$$\bar{z}_{rail}^{Normal} = g'(\bar{y}^{Normal}) + z_{rail}(Idx_{contact_k}(ini)) \quad (7b)$$

$$\bar{r}^{Normal} = h'(\bar{y}^{Normal}) + r(Idx_{contact_k}(ini)) \quad (7c)$$

where f' , g' , and h' are the interpolation functions in normal direction for the wheel, rail respectively wheel radius.

2.5 Normal contact problem

The normal contact problem can now be solved in the local coordinate system for each patch. It is solved by calculating for each patch k , the equivalent penetration, the patch shape and centroid, and finally the normal force.

2.5.1 Equivalent penetration

The equivalent penetration (or approach) is determined by the artifact of the equivalent penetration circle segment. The shape of the interpenetration region (S) is simply calculated as the difference

between the wheel profile and the rail profile in normal direction, while the width (w) of the contact patch is determined from the difference between the maximum and minimum lateral coordinates of the contact patch in normal direction,

$$S(\bar{y}^{Normal}) = \bar{z}_{wheel}^{Normal} - \bar{z}_{rail}^{Normal}, \quad w = \Delta \bar{y}^{Normal}. \quad (8)$$

The shape function area,

$$A_s = \int_{\bar{y}^{Normal}(ini)}^{\bar{y}^{Normal}(end)} S dy, \quad (9)$$

will have to be to the area of the equivalent circle segment. The equivalent contact method now uses w and A_s to calculate the equivalent penetration as the height of the circle segment that has the same area and width of the found patch,

$$A_c = \frac{1}{2} \frac{w^2}{4} \left(\frac{\beta - \sin(\beta)}{\left(\sin\left(\frac{\beta}{2}\right)\right)^2} \right), \quad (10)$$

where, β is the circular segment angle. Find the angle β by minimizing the problem,

$$A_s - A_c = 0. \quad (11)$$

Many libraries are now available to solve this kind of problems numerically. In SimRail the ‘fsolve’ function is used. This is present in most libraries (C, C++, Python, Matlab). Once the angle β is found, it is possible to calculate the circumference radius (R_c) and the height (therefore the penetration δ) of the circular segment,

$$R_c = \frac{w}{2 \sin\left(\frac{\beta}{2}\right)} \quad (12a)$$

$$\delta = 2R_c \left(\sin\left(\frac{\beta}{4}\right) \right)^2 \quad (12b)$$

The procedure result is shown in the example of Figure 4, where the shape of S is compared with the equivalent circle with the same area. The equivalent penetration method proposed in [1] now takes the penetration δ and increases it to find the equivalent penetration used to calculate the normal force:

$$\delta_{eq} = \frac{\delta}{0.55} \quad (13)$$

2.5.2 Patch Shape and Centroid

According to Piotrowski and Kik [2] the contact patch is symmetric with respect to the longitudinal plane. Therefore, it is sufficient to calculate the leading edge shape of the contact. To do so it is necessary to calculate the local rolling radius r_{local} . According to the Handbook of Railway vehicles [3],

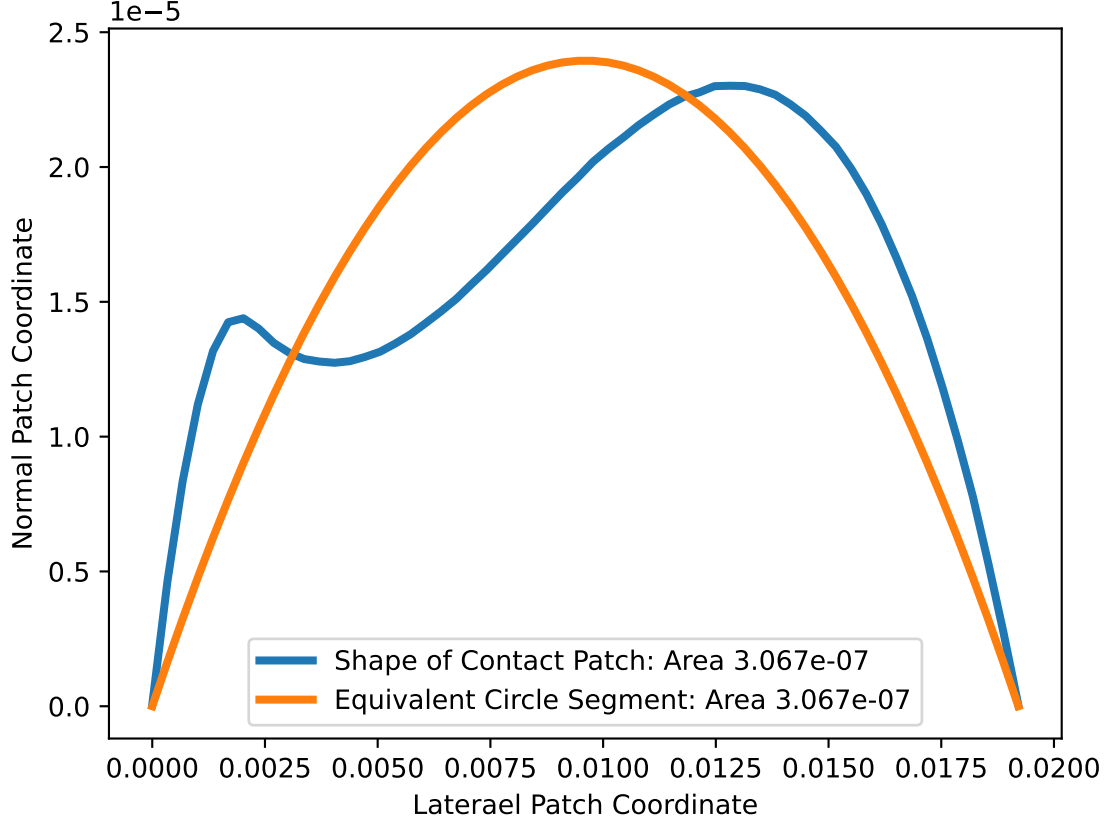


Figure 4: Contact Patch and Equivalent Circle Area

$$r_{local} = \frac{\bar{r}^{Normal} \left(\frac{end}{2} \right)}{\cos(\alpha)} \quad (14)$$

where, $\bar{r}^{Normal} \left(\frac{end}{2} \right)$ represents the middle point of the normal radius vector, and α is the contact rotation angel defined in eq. 4. The leading edge of the contact patch can now be determined as,

$$x_l = \sqrt{2r_{local}S}. \quad (15)$$

The equivalent ellipse semi axes can be simply determined,

$$a = \max(x_l), \quad b = \frac{w}{2}. \quad (16)$$

The visual comparison between the shape of the contact patch determined by eq. 15 and the equivalent ellipse is shown in Figure 5. Giving the symmetry assumption of the Piotrowski and Kik approach, the point at which normal and tangential forces will be applied can be identified by the contact patch the centroid,

$$C_y = \frac{\int_{\bar{y}^{Normal}(ini)}^{\bar{y}^{Normal}(end)} (x_l \cdot \bar{y}^{Normal}) dy}{\int_{\bar{y}^{Normal}(ini)}^{\bar{y}^{Normal}(end)} x_l dy}. \quad (17)$$

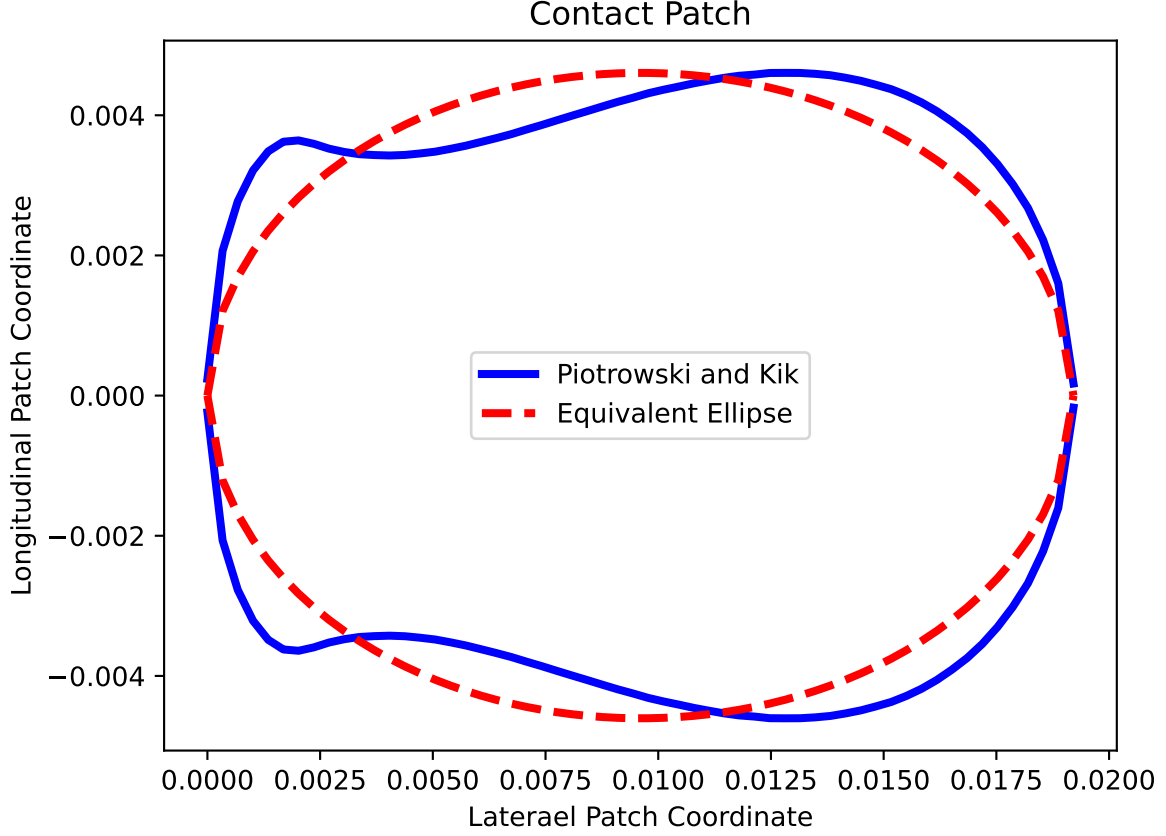


Figure 5: Contact patch determined with the interpenetration shape vs equivalent ellipse contact patch

2.5.3 Calculate Normal Force

At this stage the Hertz contact theory is used to calculate the normal force F_c using the elliptical dimensions a and b , and the determined equivalent penetration δ_{eq} . First calculate the A and B coefficients to find the contact angle Θ ,

$$A = \frac{\delta_{eq}}{a^2}, \quad B = \frac{\delta_{eq}}{b^2}, \quad \Theta = \arccos\left(\frac{A - B}{A + B}\right) \cdot \frac{180}{\pi}. \quad (18)$$

The normal force can now be evaluated,

$$N = \frac{4}{3} E^* \sqrt{\frac{\left(\frac{\delta_{eq}}{\Gamma(\Theta)}\right)^3}{A + B}}, \quad (19)$$

where,

$$E^* = \left(\frac{(1 - \nu^2)}{E} \right)^{-1}, \quad (20)$$

assuming the wheel and rail materials are isotropic, where E^* is the equivalent elastic modulus, E is the elastic modulus of the material, and ν is the Poisson's ratio. The coefficient function $\Gamma(\Theta)$ has been determined starting from the values obtained from the software SIMPACK reversing

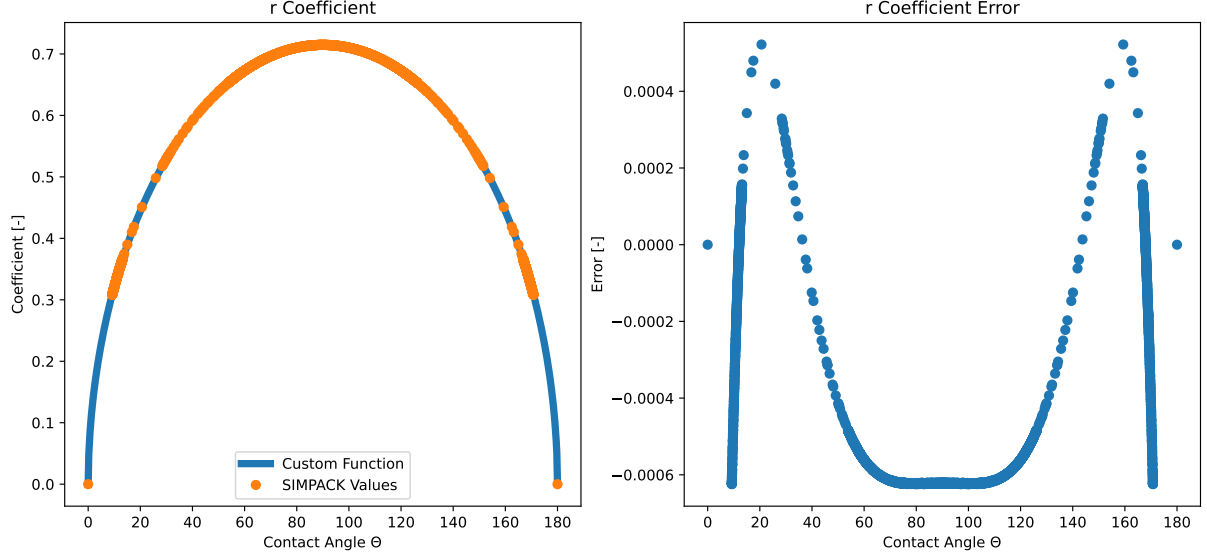


Figure 6: $\Gamma(\Theta)$ coefficient values and error

the equation defining the normal force N . An interpolation function is obtained starting from an ellipsoid shape with semiaxis $\Theta = 90^\circ$. The ellipsoid shape has been adopted starting from observation of the values given in Handbook [3] for the same coefficient,

$$\Gamma(\Theta) = \left(\sqrt{1 - \frac{(\Theta - 90)^2}{90^2}} \right)^{C_1} \cdot \frac{1}{C_2}, \quad (21)$$

where, C_1 and C_2 are obtained by minimizing the error between the points obtained from SIMPACK and the points obtained with the function. The two coefficients have the following values,

$$C_1 = 1.026600611713163413, \quad C_2 = 1.397839912270349316. \quad (22)$$

The coefficient function $\Gamma(\Theta)$ is shown in Figure 6 together with the error for each point obtained from SIMPACK. The approximated function has a R^2 value of 0.99999 and a maximum absolute error of 6.25e-4. During the fitting phase, $\Gamma(\Theta)$ was enforced to pass through the points $(\Theta = 0^\circ, \Gamma(\Theta) = 0)$ and $(\Theta = 180^\circ, \Gamma(\Theta) = 0)$.

2.6 Re-colocation of the results on the profiles coordinates

The normal force and the centroid (point at which the normal force will be applied) is found in the local normal coordinate system. Therefore, to apply the solution to the rail and wheel coordinate system, the reverse steps of paragraph 2.4 are needed. First, find by interpolation the vertical normal coordinate of the centroid for the wheel and the rail $\bar{z}_{wheel}^{Normal}(C_y)$, $\bar{z}_{rail}^{Normal}(C_y)$. Now it is possible to calculate the location of the centroid on the wheel and rail profile for each contact patch found.

$$\begin{bmatrix} y_{rail}(C_y) \\ z_{rail}(C_y) \\ 0 \end{bmatrix} = R_\alpha^{-1} \begin{bmatrix} \bar{z}_{rail}^{Normal}(C_y) - z_{rail}(Idx_{contact_k}(ini)) \\ C_y \\ 0 \end{bmatrix} + \begin{bmatrix} y_{rail}(Idx_{contact_k}(ini)) \\ z_{rail}(Idx_{contact_k}(ini)) \\ 0 \end{bmatrix} \quad (23a)$$

$$\begin{bmatrix} y_{wheel}(C_y) \\ z_{wheel}(C_y) \\ 0 \end{bmatrix} = R_\alpha^{-1} \begin{bmatrix} \bar{z}_{wheel}^{Normal}(C_y) - z_{wheel}(Idx_{contact_k}(ini)) \\ C_y \\ 0 \end{bmatrix} + \begin{bmatrix} y_{rail}(Idx_{contact_k}(ini)) \\ z_{wheel}(Idx_{contact_k}(ini)) \\ 0 \end{bmatrix} \quad (23b)$$

It is possible to calculate the vertical and lateral components of the normal force in the wheel rail coordinates system, that is absence of components of creep forces can be considered Q and Y forces, simply by,

$$Q = N \cos(\alpha), \quad Y = N \sin(\alpha). \quad (24)$$

3 Results

The developed routine can now be tested against SIMPACK results. To do so the static contact for the wheel moving relatively to the rail location in lateral direction is considered. Here, the wheel is moved from -20 to 20 mm with a step of 0.08 mm. The vertical position of the wheel with respect to the rail is calculated to balance an arbitrary imposed Q force. Therefore, for each lateral position of the wheel with respect to the rail, an equivalent penetration is found that generates a normal force that balance in vertical direction the imposed Q force. Setting Q as 50 kN and applying the procedure gives the results of Figure 7, Figure 8, and Figure 9. This test has been performed with wheel profile S1002 and rail profile UIC60 with rail inclination 1:40, 1:30, and 1:20 respectively. The gauge is 1.435 m, $b_0 = 0.75$ m, $E = 210$ MPa, and $\nu = 0.28$. The good agreement of the results can be quantified as in Table 1, where R^2 , the maximum absolute error, and the minimum absolute error are reported for the key values considered and the three rail inclination. In Figure 10 the count of the contact patches for the three considered cases are compared between SIMPACK and the reconstructed results. The results show that the implemented method in SimRail is able to reproduce the results of SIMPACK with a very high accuracy.

Table 1: Errors of reconstructed values with respect to SIMPACK

Value	R^2			Max Abs Error			Min Abs Error		
Rail Inclination	1:40	1:30	1:20	1:40	1:30	1:20	1:40	1:30	1:20
Equivalent Penetration [mm]	0.991	0.999	0.995	5.91e-2 @ $y = 5.33$	7.64e-3 @ $y = 6.61$	2.25e-2 @ $y = 6.61$	1.85e-6 @ $y = -0.12$	1.58e-6 @ $y = 1.88$	3.84e-5 @ $y = -12.22$
Normal Force [kN]	0.990	0.997	0.992	40.52 @ $y = 5.33$	18.26 @ $y = 6.69$	32.65 @ $y = 6.61$	1.14e-5 @ $y = 3.09$	7.29e-5 @ $y = -19.68$	6.98e-6 @ $y = 5.65$
a/b ratio [-]	0.999	0.999	0.998	2.42 @ $y = 5.33$	1.34 @ $y = 6.69$	2.51 @ $y = 6.61$	2.20e-4 @ $y = 2.61$	1.27e-4 @ $y = 4.05$	3.23e-5 @ $y = -11.02$
Patch Length [mm]	0.992	0.995	0.987	7.46 @ $y = 5.33$	5.86 @ $y = 6.69$	9.13 @ $y = 6.61$	6.69e-5 @ $y = -9.02$	1.37e-5 @ $y = -6.85$	4.25e-7 @ $y = 7.89$
Patch Width [mm]	0.997	0.999	0.980	3.44 @ $y = 5.33$	1.47 @ $y = 6.53$	5.03 @ $y = 6.13$	1.14e-4 @ $y = -13.43$	3.64e-5 @ $y = 9.42$	2.43e-5 @ $y = -8.7$
Patch Area [mm ²]	0.992	0.995	0.969	38.77 @ $y = 5.01$	42.41 @ $y = 6.61$	42.13 @ $y = 6.53$	3.69e-4 @ $y = -5.01$	6.07e-4 @ $y = -19.76$	7.57e-4 @ $y = -8.94$
Contact Position Wheel [mm]	0.999	0.999	0.999	9.04 @ $y = 5.33$	1.82 @ $y = 2.93$	1.46 @ $y = 6.53$	6.38e-4 @ $y = 3.09$	1.14e-4 @ $y = -13.43$	3.64e-5 @ $y = 9.42$
Contact Position Rail [mm]	0.999	0.999	0.999	9.04 @ $y = 5.33$	1.82 @ $y = 2.93$	1.46 @ $y = 6.53$	6.38e-4 @ $y = 3.09$	1.14e-4 @ $y = -13.43$	3.64e-5 @ $y = 9.42$

4 Conclusion

The SimRail library has been developed for the calculation of the normal contact problem with the equivalent elastic method. The step by step implementation of the method has been described and the results have been compared with the results of SIMPACK. The results show that the implemented method in SimRail is able to reproduce the results of SIMPACK with a very high accuracy. The library is available on GitHub (<https://github.com/RoccoLiberoGiossi/SimRail>) and can be used for further research and development in the field of railway vehicle dynamics. The available library also implements the Piotrowski and Kik method to calculate the normal force. A complete example of use is available in the repository.

References

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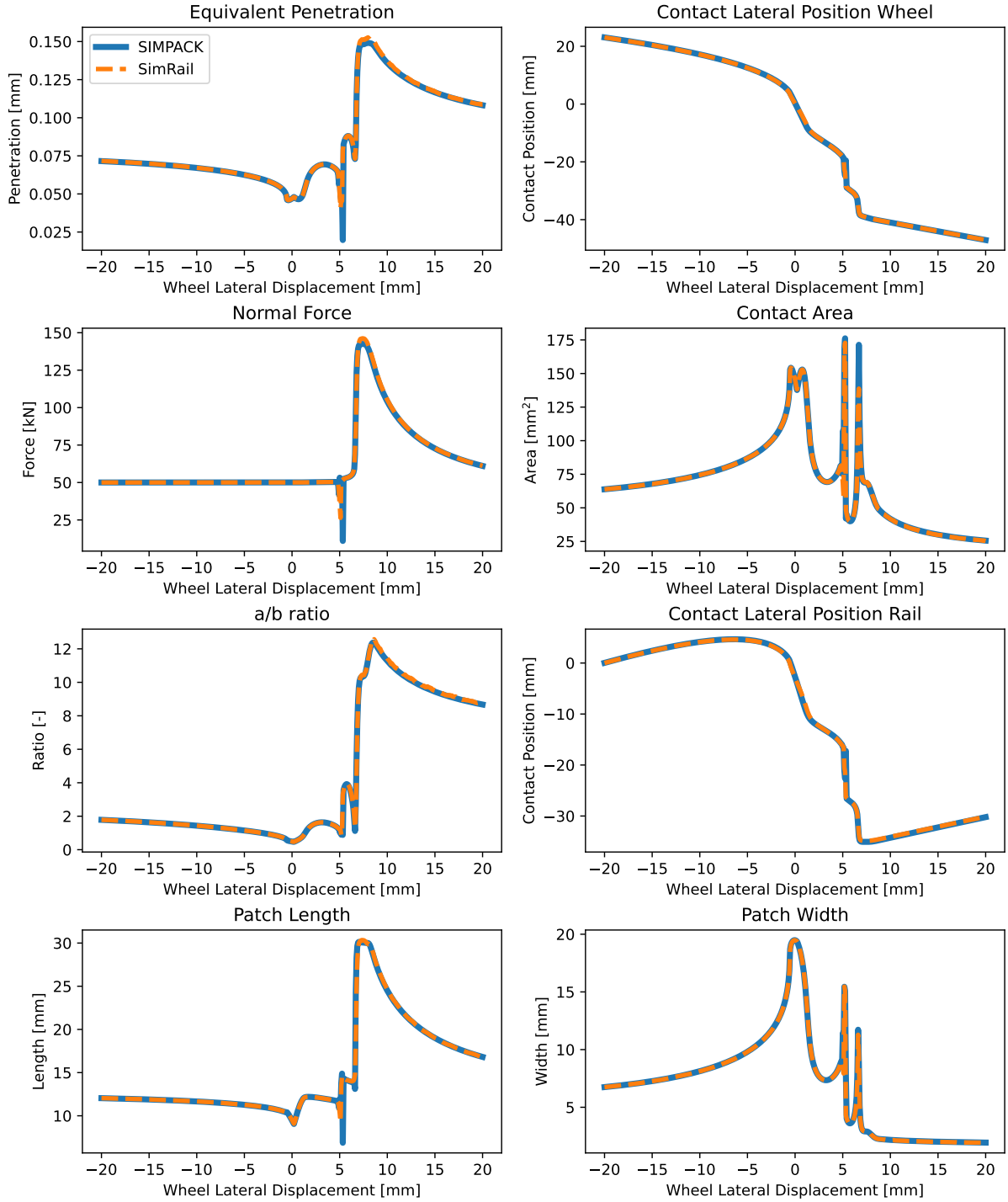


Figure 7: Results of contact search for rail inclination 1:40

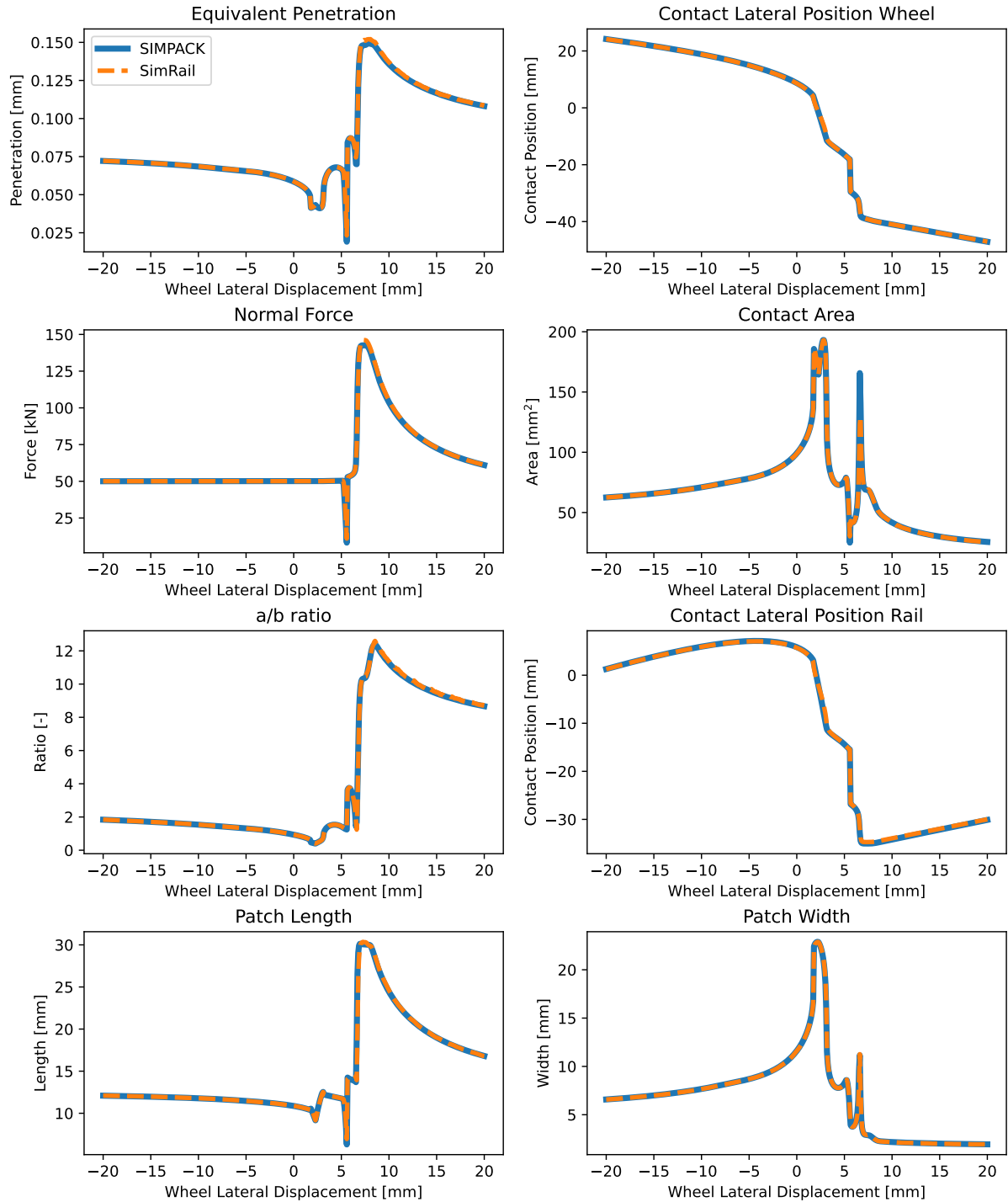


Figure 8: Results of contact search for rail inclination 1:30

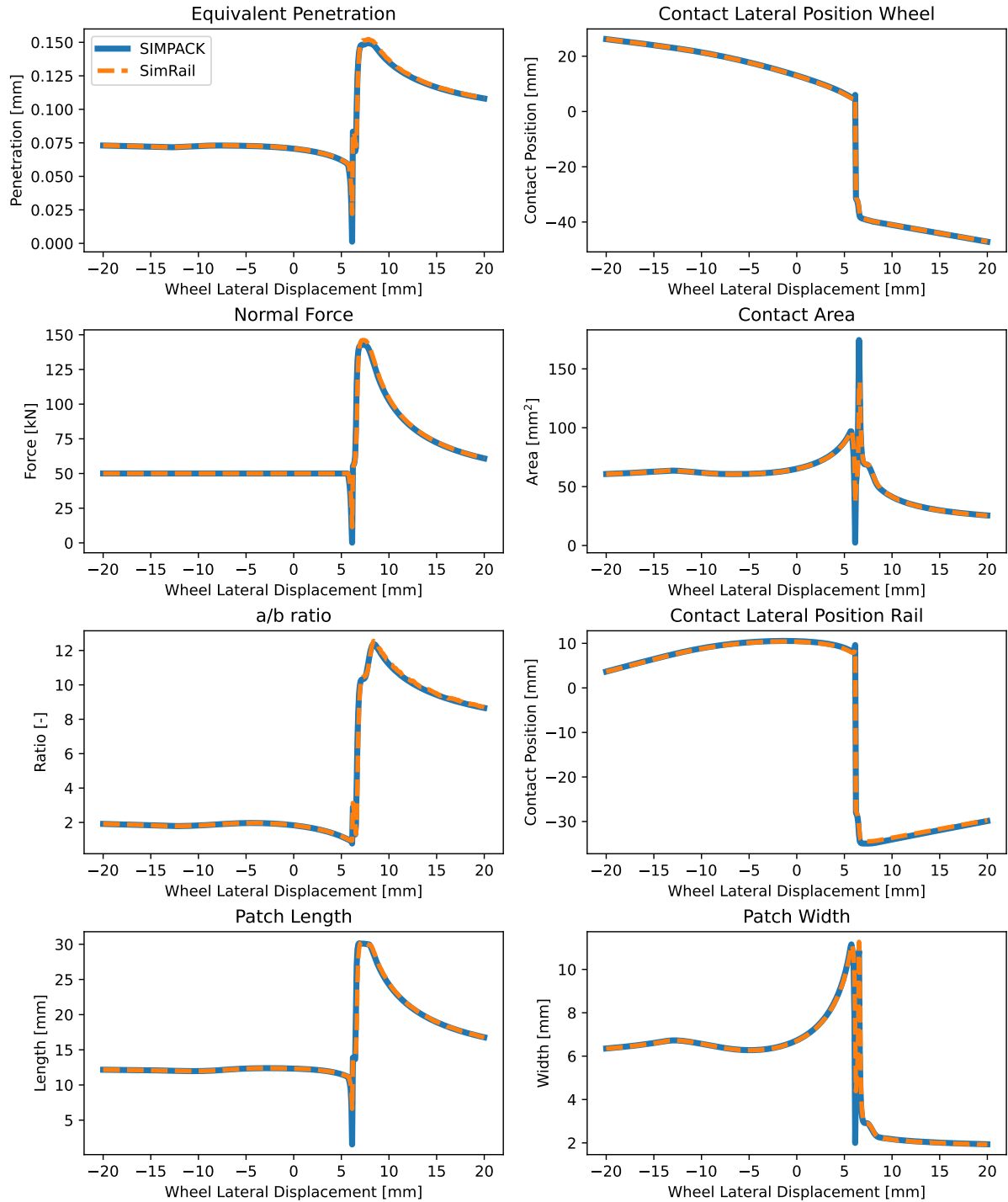
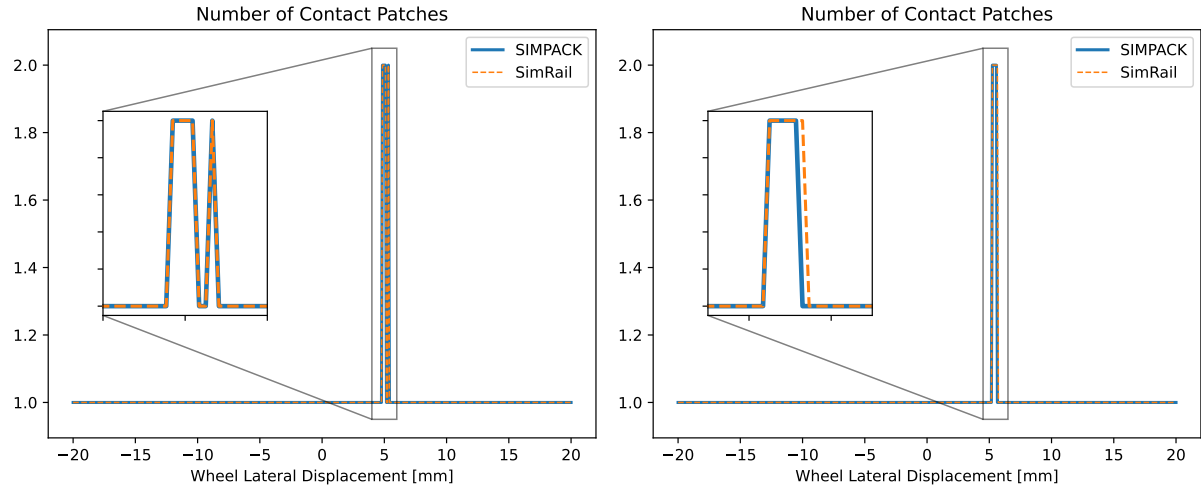
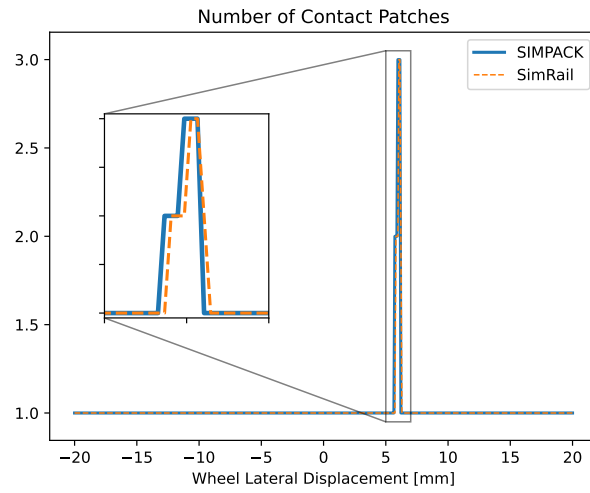


Figure 9: Results of contact search for rail inclination 1:20



(a) Rail inclination 1:40

(b) Rail inclination 1:30



(c) Rail inclination 1:20

Figure 10: Number of contact patches as function of lateral movement of the wheel